

University of Colombo School of Computing (UCSC)

University of Colombo

SCS 1307 – Probability and Statistics

Assignment 01

Semester I – 2026

Index Number: 25000799
25001078 – Temp

Instructions:

1. Answer all questions. Each question carries 25 marks
2. Show all necessary steps in your calculations
3. Duration: 1 Hour

Q1. Write answers for the questions below.

- (i) A password is formed by selecting one digit from 0 to 9 and one letter from A to Z. Assuming 0 is an even digit, what is the probability that the digit is even and the letter is a vowel? [8 marks]
- (ii) A system generates two random integers from 1 to 6 with replacement. What is the probability that their sum is at least 9? [8 marks]
- (iii) In a class, 70% of students use Python, 45% use R, and 30% use both Python and R. Given that a student uses Python, what is the probability that the student also uses R? [9 marks]

Q2. A university network records user activity. Let

- A = event that a student uses the university Wi-Fi on a given day.
- B = event that a student accesses the Learning Management System (LMS) on that day.

Suppose $P(A) = 0.7, P(B) = 0.5, P(B | A) = 0.6$

- (i) Find $P(A \cap B)$. [6 marks]
- (ii) Find $P(A \cup B)$. [6 marks]
- (iii) Find $P(A | B)$. [8 marks]

(iv) Determine whether events A and B are independent. Justify your answer. [5 marks]

Q3. A university stores student data on one of three servers: A , B , and C .

- 40% of requests are processed by Server A .
- 45% of requests are processed by Server B .
- 15% of requests are processed by Server C .

The probability that a request fails is:

- 0.03 for Server A
- 0.05 for Server B
- 0.08 for Server C

(i) What is the probability that the next request is processed by Server B and does not fail? [8 marks]

(ii) What is the probability that the next request fails? [8 marks]

(iii) Given that a request has failed, find the probability that it was processed by Server C . Give your answer to three decimal places. [9 marks]

Q4. A web server receives requests independently. The probability that a request is invalid is 0.15. Suppose 10 requests are received.

Let X be the number of invalid requests.

(i) State the distribution of X with its parameters. [5 marks]

(ii) Find the probability that exactly 2 requests are invalid. [10 marks]

(iii) Find the mean and variance of X . [10 marks]

..... End